

RADWIN

TerraNet Series User Manual

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1 Introduction

1.1 Document Scope

This document describes how to configure and manage the TerraNet Outdoor Units (ODUs). It describes the V40 and V120 models, concepts of operation, provides technology overview, and troubleshooting, as described in the following sections:

[Connecting a Laptop to the ODU](#)

[Viewing Devices and Link Status](#)

[Configuring the ODU](#)

[Configuring Service Parameters](#)

1.2 Revision History

	Date	Document Revision	SW Release	Revision details
1.	March 20 th	1.0	1.0.00	Initial Revision
2.	June 24 th	2.0	1.0.00	Few updates
3.	July 25 th	3.0	1.0.00	Includes Web UI changes
4.	August 5 th	4.0	1.0.00_b54561	Final release
5.	February, 13 th	4.1	1.13.00_b54597	Add V90 support
6.	September, 8th	4.2	1.14.1_b54664	New SW release supporting Japan regulation

1.3 Product Family Overview

RADWIN TerraNet family delivers up to 2.3Gbps aggregate capacity in Point-to-Multi-Point or Point-to-Point architecture and is ideal choice for smart cities in LOS and short-range conditions.

The TerraNet portfolio includes the following models:

RADWIN 2000-E Family Model Comparisons

Model Name	P/N	Product Name	Max Throughput	Form Factor
V120	RW-6V02-9160	RW6000/V120/F60/UNI/INT	2.3Gbps	Integrated
	RW-6V02-I160	RW6000/V120/F60/LIM/INT		
	RW-6V02-J160	RW6000/V120/F60/JPN/INT		
V40	RW-6N02-9160	RW6000/V40/F60/UNI/INT		Integrated
	RW-6N02-I160	RW6000/V40/F60/LIM/INT		
	RW-6N02-J160	RW6000/V40/F60/JPN/INT		
V90	RW-6P02-9260	RW6000/V90/F60/UNI/INT		Integrated and External Antenna Lenses
	RW-6P02-I260	RW6000/V90/F60/LIM/INT		
	RW-6P02-J260	RW6000/V90/F60/JPN/INT		

The TerraNet V90 can be used with external antennas, for more information about how to mount these external antennas on V90, please refer to the Installation guide. Here is a short description of this set of antenna lenses:

P/N	Product Name	Azimuth Width (°)	Elevation Width (°)	Antenna Gain (dBi)
RW-9130-5771	RW-ANT/5771/LENS/30/45	45	10	30
RW-9230-5771	RW-ANT/5771/LENS/30/90	90	10	30
RW-9333-5771	RW-ANT/5771/LENS/33	6	6	33
RW-9437-5771	RW-ANT/5771/LENS/37	4	4	37



Some options and models may not be available in your region due to regulatory compliance.

RADWIN TerraNet products are compliant with EN IEC 62368-1:2020+A11 safety standard.

1.4 Technology Overview

1.4.1 Higher Capacity - Over Longer Distances

TerraNet Product Portfolio leverages the cutting-edge 60GHz (802.11ad) technology, relying on techniques RADWIN market proven solutions to bring performance to a new level.

Extended Frequency Range (Channels 1-6)

With the ability to use the entire 60GHz band (57-71GHz), TerraNet supports channels 5 and 6 (where enabled by regulation). Channels 5 and 6 have a much lower oxygen absorption than in channels 1 to 4. This capability leads to better performance at longer range.

Advanced Beamforming Antenna

TerraNet V120 and V40 models are both equipped with advanced 3D beamforming antenna. This cutting-edge solution enables fast automatic link establishment and accurate alignment in less than 1 minute, with continuous beam tracking and performance optimization of active links.

A Beam Steering algorithm is used to allow auto-alignment with wide scanning range for both azimuth and elevation beams in any operation mode (Access Point, Station, PTP unit).

Automatic Modulation and Coding

RADWIN proprietary algorithm for Automatic Adaptive Rate adjusts the Modulation and Coding Scheme and continuously checks for potential better MCS without affecting the current level of service.

TerraNet product family supports the following modulation schemes: 16QAM, BPSK, QPSK with the following error correction code rates 1/2, 5/8, 3/4, 7/8 and 13/16.

The automatic adaptive rate maximizes Ethernet throughput while ensuring a minimum Error Rate, enhanced robustness and better performance under interference.

TerraNet performs rate optimization independently for uplink and downlink.

Non-Interrupted Transmission

The non-Interrupted transmission technique keeps transmissions regardless of changing conditions in the channel, leaving the on-the-fly corrections to operate while the communication flows remain stable and robust.

Adaptive DL/UL Ratio

TerraNet family supports adaptive DL/UL ratio. This capability allows dynamic capacity optimization allocated according to traffic demands.

Half/Full Channel Bandwidth

TerraNet can be configured to utilize a “full” 2160 MHz channel or in “half” 1080 MHz channel bandwidth.

Security and Encryption

The air interface supports advanced AES-GCMP 128-bit authentication and over-the-air encryption. Link password is required to establish a link.

1.5 Modes of Operation

Any TerraNet product can be configured for Point-to-Multi-Point and Point-to-Point operation modes.

1.5.1 Point-to-Multi-Point

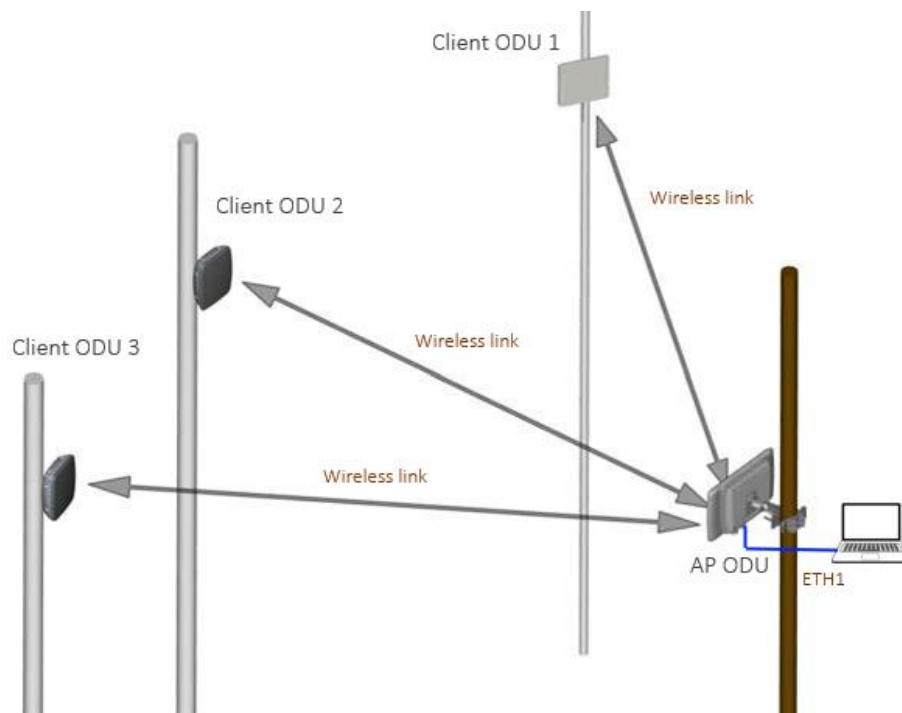


Figure 1: PtMP Connection Scheme

Any TerraNet ODU can be configured either as Access Point (AP) or as a Station (Client). In Point-to-Multi-Point Access Point mode, up to 32 Stations can be connected.

1.5.2 Point-to-Point

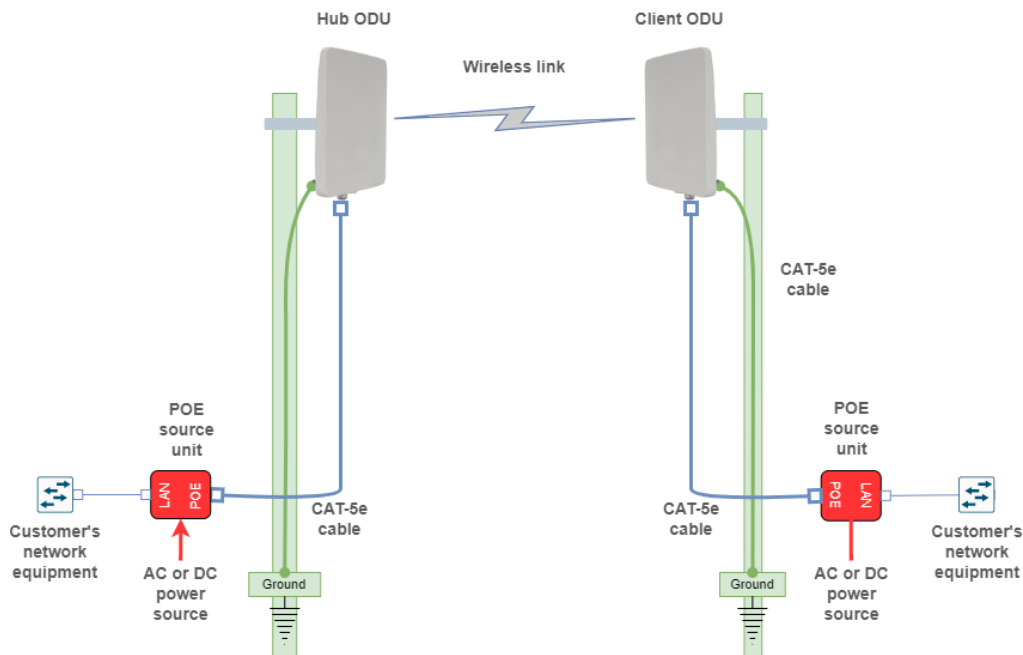


Figure 2: PtP Connection Scheme



For simplicity, Lightning Protection Units (LPUs) are not shown in the following figure but are recommended.

Any TerraNet ODU in a PTP link can be configured either as PTP Access Point or as PTP Station.

Both for PTMP and for PTP, an Access Point will transmit a beacon on the selected frequency channel, and a Station will scan for the beacon and connect to the Access Point.



Radwin recommends that any ODU connected to the operator's backbone network will be configured as an Access Point, while ODU's at remote sites (closer to the end customer) will be configured as a Station.

1.5.3 Link Establishment and Starting Service

ODU Activation and Initial Setup

An ODU comes out of the box configured by default as a Station. You can change the ODU configuration between AP mode/PtP Master and Client through the Web UI.

Connecting to the Outdoor Unit for the first time, you need to connect an ethernet cable into either the ETH0 (2.5G) or the ETH1 (1G) port which will connect it to the radio.

By default, DHCP Station is enabled on the main local network bridge. If your device cannot get an IP from an upstream DHCP server, it will be fallback to 10.0.0.120 automatically.

The device is also accessible through the device additional "alternative local IP" of 169.254.1.1.

You should then access your device's local web UI via web browser at the DHCP-assigned IP or the fallback IP mentioned in the previous step.

Then login using the default login credentials of username and password.



You will automatically be logged out of your session if you're inactive for more than 30 minutes.

To establish a link between ODUs, you need to activate the ODU by configuring essential parameters such as link ID, password, operating band and channels in the AP.

By using the default parameters, the ODUs will automatically connect one to the other after a few minutes.

2 Login & access

RADWIN

Sign In

Username

root

Password

admin

6d

Log In

Figure 3: Login – Default Username and password



The device's default fallback IP is 10.0.0.120, we recommend you change the device's default user credentials after you log in for the first time.

The session is logged out automatically if inactive for more than 30 minutes.

We recommend changing the device's default user credentials after the first log in.

3 TerraNet Web Interface

The TerraNet Web Interface provides access to six different sections described in this document:

- Dashboard: gives you access to all important information about the device on a single page
- Activity: Logs all actions, modifications and events occurring on the device
- Configuration: Used to configure the device
- Users: manages the users' credentials
- Tools: used to manage and debug the device
- Backup and Recovery: Manage system FW backups

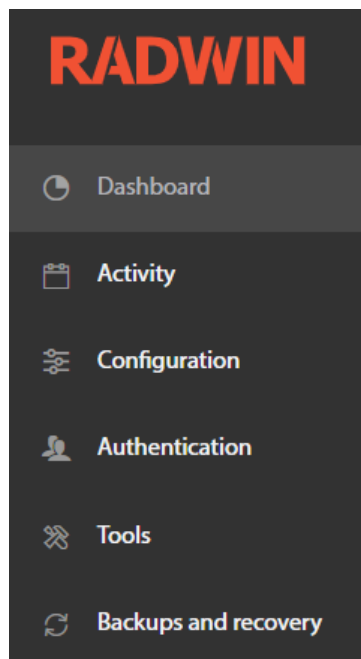


Figure 4: TerraNet Web Menu

At the bottom left part of WebUI window, you could view the firmware build information of the device by overring the mouse on the ⓘ.

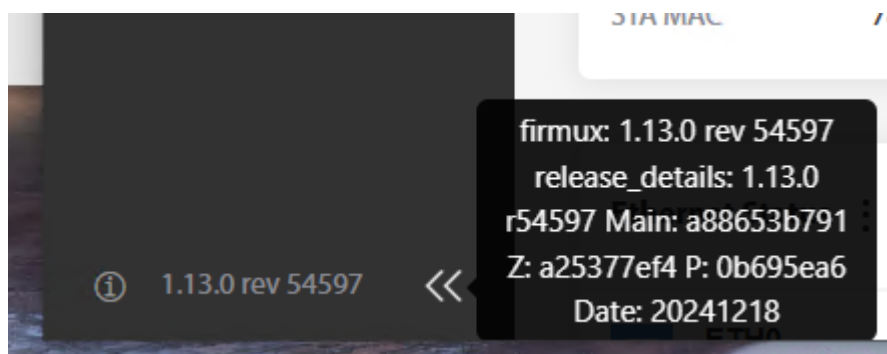


Figure 5: TerraNet Build Info

4 Dashboard

The device dashboard shows the overall status of your device, including:

- System Information.
- Wireless Status.
- Ethernet Status.
- ETH ports Throughput.
- Network Information.
- System Resources.

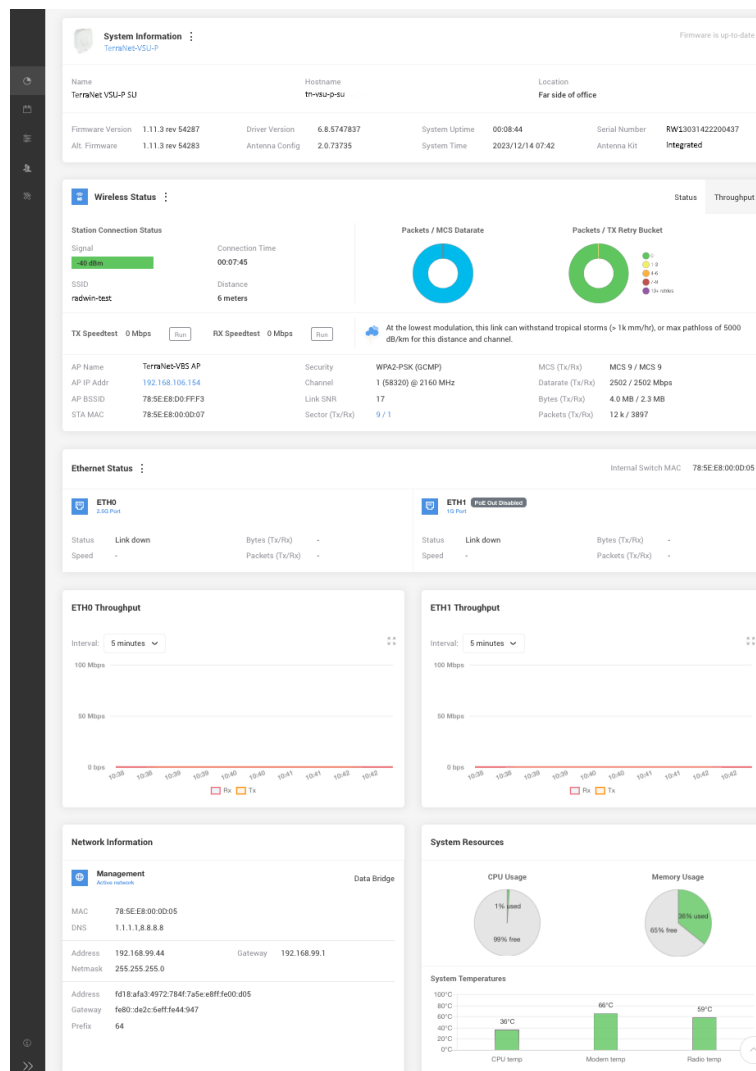


Figure 6: Dashboard

4.1 System Information

System Information 					
RW-6V02-9160					
Name		Hostname		Location	
V120		AP-V120		RADWIN PS LAB	
Firmware Version	1.0.0 rev 54545	Serial Number	P234706202400001	System Uptime	15 days 21 hours
Alt. Firmware	1.0.0 rev 54544	Driver Info	7.1.6800270 / 1.0.110079	System Time	2024/07/11 10:01

Figure 7: System Information

This section displays information about your device, including:

- Device name, location, and hostname:
 - Device name is used to populate LLDP discovery fields
 - Hostname is used when sending DHCP requests
- The firmware versions running on each device boot bank
 - Firmware version: active firmware currently running
 - Alt. firmware: alternate/backup firmware version (will be used in case active firmware fails to boot)
- Driver and antenna config versions: these fields are mainly used for debugging purposes.
- Current system time: This is the device's current date and time. Time zone and other time settings are configurable from the System settings page.
- Antenna kit: On the current TerraNet portfolio, the antenna is integrated.
- Device uptime since last reboot.

4.2 Wireless Status

The wireless status differs depending on the device configuration:

- Station or point-to-point Station mode
- Access Point mode or point-to-point Access Point mode

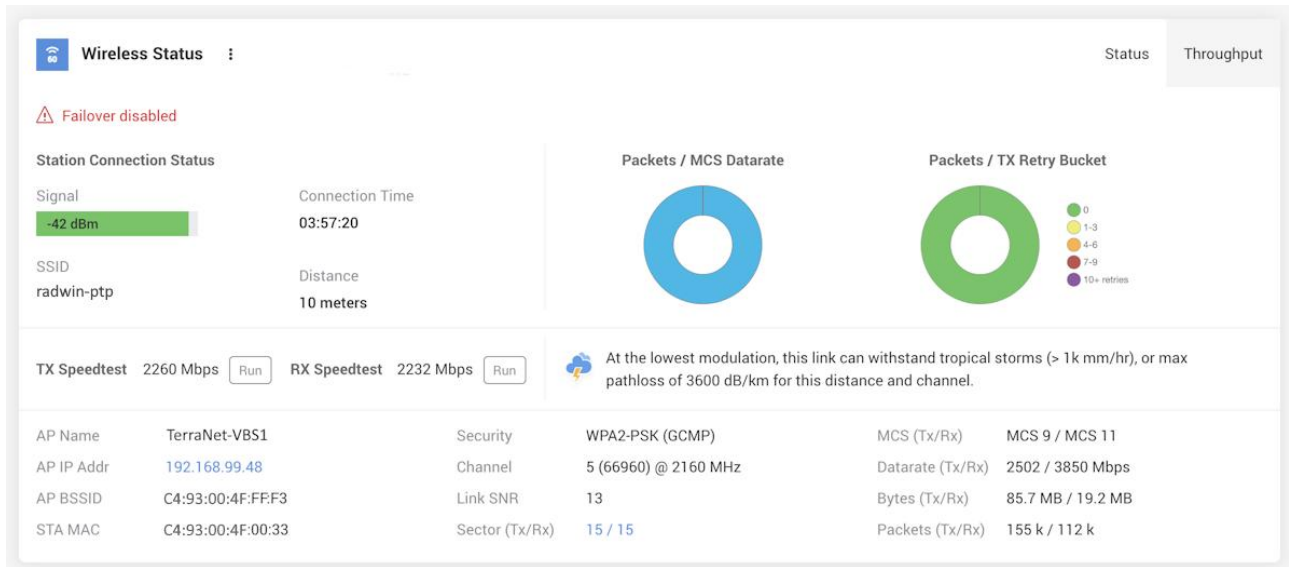


Figure 8: Wireless Status – Station Mode

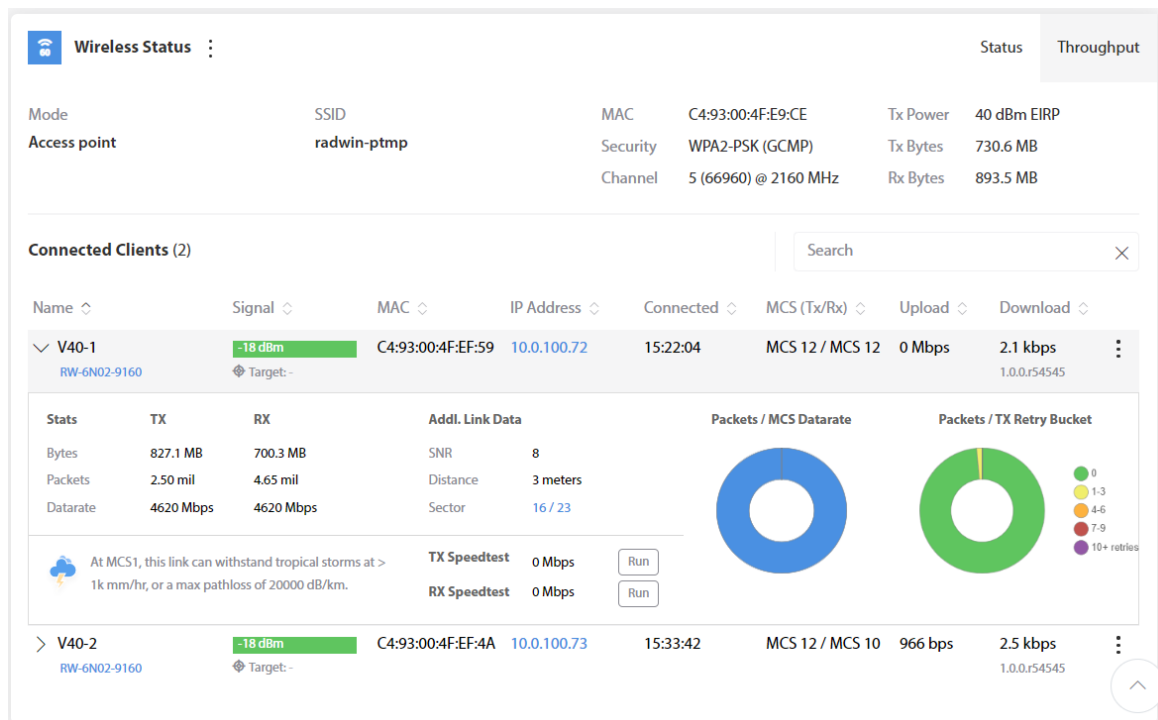


Figure 9: Wireless Status – AP Mode



In Point-to-Point configuration, on the AP side, you will only have a single client connected.

4.2.1 Disconnect a Station

You can “kick off” an associated Station from a device operating as AP by clicking the "three dot" menu at the right end of the row and selecting "Kick client". This can be useful when using station connection profiles to force your station to connect to a different AP.

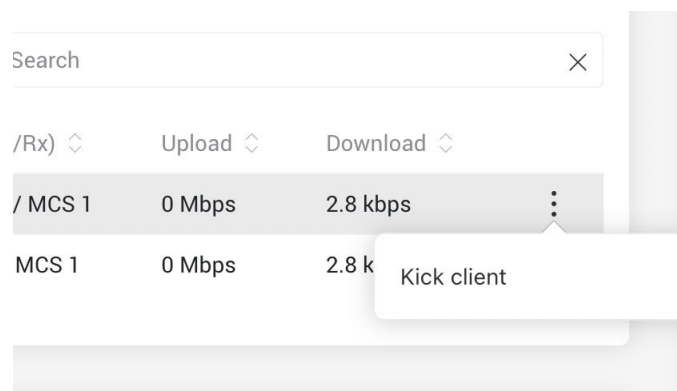


Figure 10: Wireless Status – Disconnect a Station

4.2.2 Wireless Traffic Graph

To view the wireless throughput graph, click on the upper right corner of the wireless widget on the dashboard.

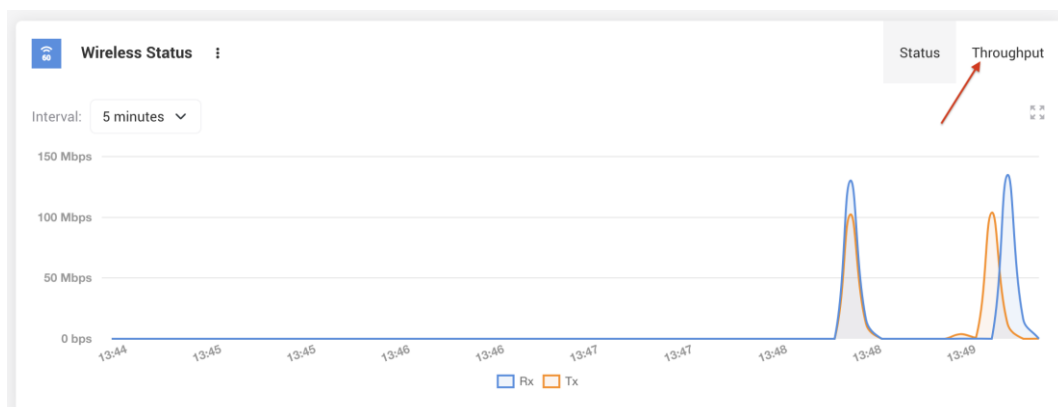


Figure 11: Wireless Status – Traffic Graph

4.2.3 Connected Clients Statistics

Through the Wireless Status tab, you can get statistics about each Client connected to an AP for both mode PtP or PtMP.

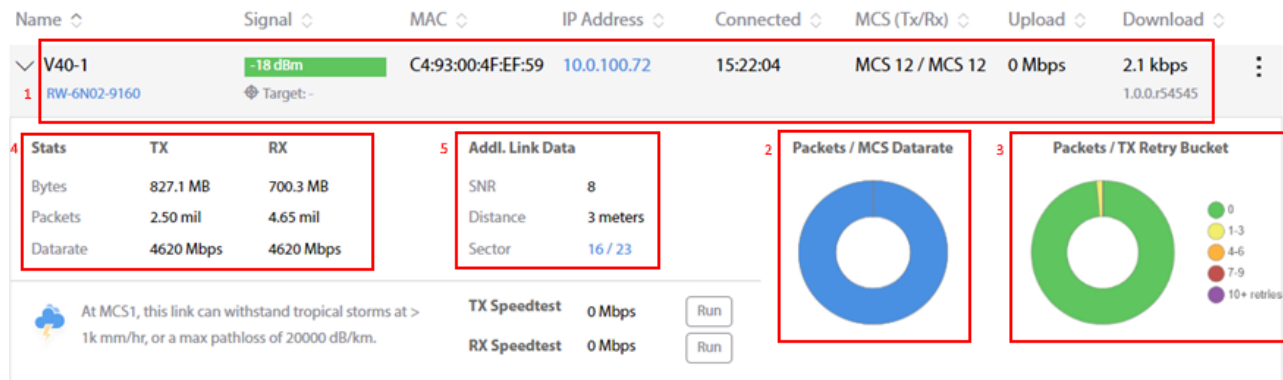


Figure 12: Wireless Status – Client Stats

- IP Address and peer name:** A connected device's management IP is available once discovery/LLDP data is available for the peer, which may take a few minutes after association. If the discovery tool or LLDP server are disabled on either the AP or station, then discovery data (including IP and device name) will not be available.
- Target RSSI:** This is an estimate of the best-case RSSI for the link, given the device models detected on either side of the link, as well as the link distance. Actual RSSI may differ from target depending on environment factors, beam location, etc... This field is not populated if the link distance is under 100 meters, or if the model of the device is not recognized.
- Packets/MCS Data rate chart:** The system observes a client's traffic during the 5-minute interval and determines how many packets were sent for each MCS rate. Lower MCS rates have a lighter yellow color, and gradually turn a darker blue the higher the rate. When clients first connect, you'll see that lower MCS rates are being used, and then when traffic increases over the link, the number of packets in the upper MCS rates will grow.
- Packets/ TX Retry bucket chart:** The system also checks to see how many packets were retried across the wireless link during the observational period and categorizes them into buckets based on the number of retries. If you have many packets outside of the "green" bucket, check your link for obstructions. If there are none, you might have an environmental issue with reflections.
- Link Availability:** This section shows how much rain (mm/h) the link can withstand at MCS1 before it goes down.
- Sector IDs:** Displays the Tx/Rx sector numbers. These numbers could be identical or different. A connected peer's sector ID can be visualized in the Sector Info Tool. These numbers identify the Tx/Rx beams for a specific device,



Data is collected at a 5-minute interval. The connected client charts "Packets/MCS Data rate" and "Packets/TX Retry Bucket" will show up only after the client has been connected for at least 5 minutes. The data is not cumulative across the lifespan of the link, only for the previous 5-minute duration.

4.2.4 SpeedTest

You can perform a speed test for each Station connected to an AP by pressing the "Run" button near the TX or RX SpeedTest labels next to a connected Station on the AP, or from the wireless status widget of the web interface of the Station.

This can be used to test throughput between the AP and Stations, or vice versa. It could be useful also to rule out ethernet or other network issues when troubleshooting link performance.

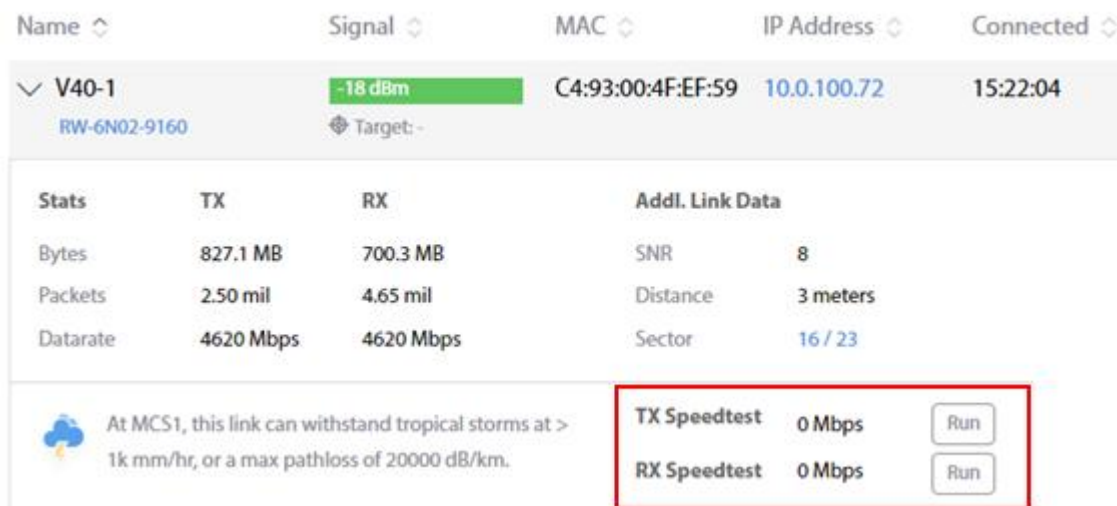


Figure 13: Wireless Status – Speedtest



The SpeedTest tool pushes small burst of packets across a link, so running it will not have any noticeable effect on customer traffic, nor on the wireless throughput graph.

5 Activity

The activity page lists recent events (last 24h), such as Station association/disassociation, user login, DHCP events, etc.

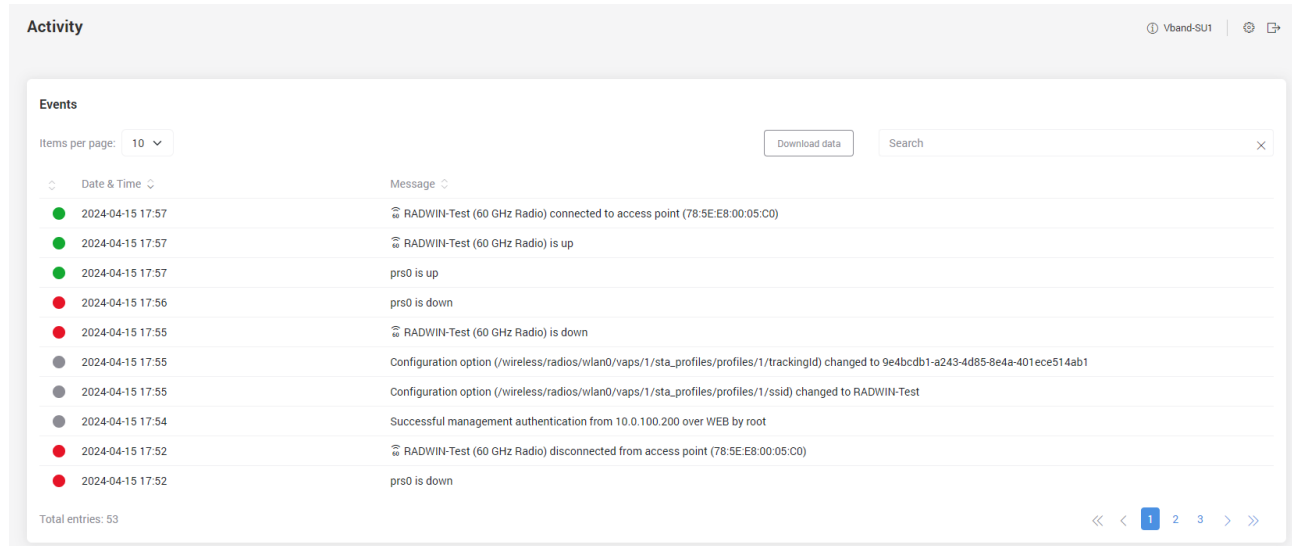


Figure 14: Activity

For the event log and other diagnostic tools, TerraNet is using the following interface names:

Interface name	Interface type	Description
Eth0	Ethernet	2.5Gbps (POE input)
Eth1	Ethernet	1Gbps (POE out)
Prs0	Wireless	PTMP Station #1 / PTP Station
----	----	----
Prs31	Wireless	PTMP Station #32

6 Configuration



Only admin-level users have rights to access and change settings on the configuration pages of the web UI.

6.1 Applying Configuration Changes

While you're configuring your device, feel free to make changes to one or more settings located on any of the configuration pages mentioned in the sections below.

Once you're done, click the Save button at the top of the page in order to write and apply the changes. Please note that your device will become temporarily unreachable while networking and other system services are restarted.

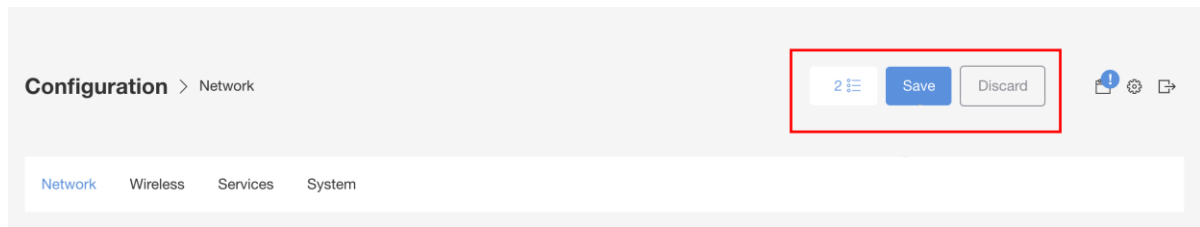


Figure 15: Configuration – Save or Discard changes

If you wish to discard your changes, refresh the page, or click the Discard button.

You can also view which changes are pending by clicking the list button to the left of the Save button.

6.2 Network Settings

The Network Settings tab under the configuration menu allows you to configure the device's ETH ports.

Different sections are available:

- Device Management
- Interface Settings
- Advanced Settings

6.2.1 Device management

Management & local networks

☐ Enable management VLAN

Management IPv4 mode

Static

Management IPv4 settings

☒ Enable IPv4 static IP

IP address

10.0.100.31

Netmask

255.255.255.0

Gateway

10.0.100.254

DNS servers

1.1.1.1

1.0.0.1

Management IPv6 settings

☐ Enable IPv6 static IP

Alternative local IP settings

☒ Enable alternative local IP

Alt. local IPv4 address

169.254.1.1

Figure 16: Device Management Settings

VLAN Management Network

Management & local networks

Warning: Management VLAN is enabled, which means that you will not be able to access this device's web UI at the management IP configured below (whether DHCP-assigned, static, or DHCP fallback), unless you are on the management VLAN network. You can still access this device natively by enabling Data Bridge Static IP or Alternative Local IP below.

☒ Enable management VLAN

Management VLAN ID

100

☒ Enable static IP on data bridge

Data bridge IPv4 address

192.168.2.1

Data bridge IPv4 netmask

255.255.255.0

Management IPv4 mode

DHCP client

Management DHCP client settings

Fallback IPv4 address

10.0.100.31

Management IPv4 netmask

255.255.255.0

☒ DHCP broadcast

☒ Custom DNS

DNS servers

1.1.1.1

Figure 17: Network – VLAN Management Network

Enable Management VLAN: enable or disable management VLAN on the device. You need to specify the VLAN ID you will use for Management VLAN. VLAN ID needs to be in the range of 2 to 4094. By default, VLAN ID is set to 100.



Once this setting is enabled, you must have your management VLAN settings correctly configured or you will not be able to reach your device again without resetting to defaults, unless you have a data bridge static IP set.

Enable static IP on data bridge: When this setting is enabled, you will be able to set an IPv4 static IP and netmask on the data bridge, giving one access to the local UI over the data network instead of the management VLAN network. This can be helpful in the case where a tech needs to have access to the

device during installation over the non-management VLAN network. Once aiming and installation is complete, this setting can be turned off, only allowing access to the web UI over the management VLAN network.

Here's an example of how a station and the devices connected to it will get an IP address assigned when operating without VLANs (Station A), with data VLAN (Station B), and with both data and management VLAN enabled (Station C):

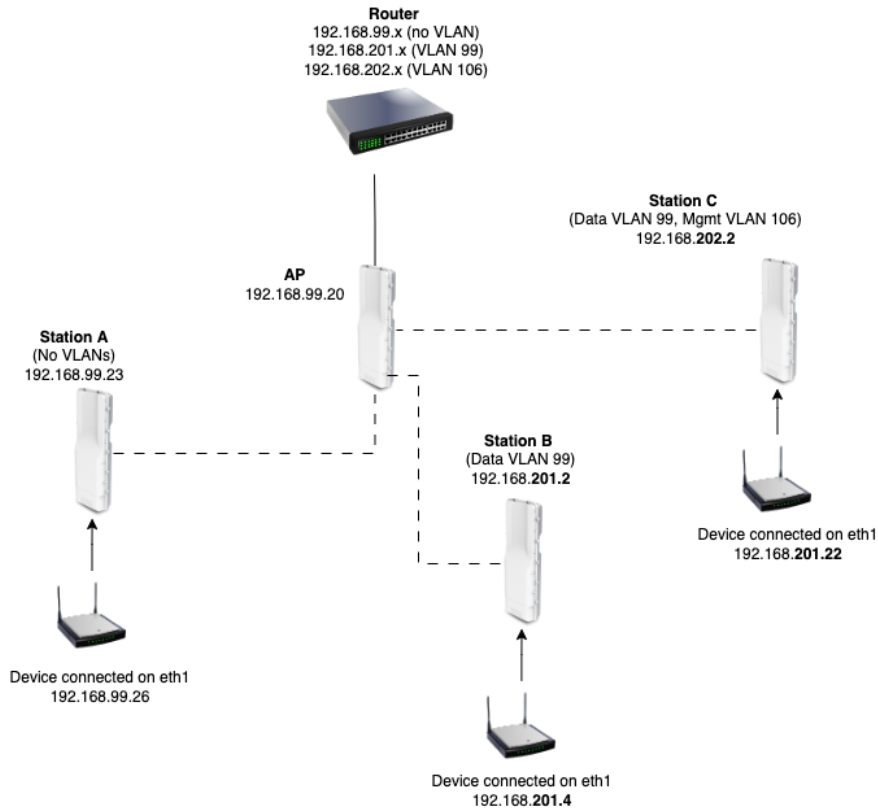


Figure 18: Network – VLAN Topology Example

Management IPv4 mode: Two modes are proposed, Static or DHCP client.

DHCP client: If you choose DHCP client, you'll have the option of setting a fallback IPv4 address and netmask, Custom DNS, and enabling DHCP broadcast (which requests DHCP broadcast replies from the DHCP server).

Static IP: If you choose Static IP as the IP mode, you will need to manually set at least one IP (IPv4 or IPv6) for the device as shown below.

Management IPv4

When Management IPv4 mode is set to static, you need to provide an IP address, a gateway and DNS Servers to the device. You could also use IPv6 instead of IPv4 or both.

Management IPv4

☒ Enable IPv4 static IP

IP address

10.0.100.31

Netmask

255.255.255.0

Gateway

10.0.100.254

DNS servers

1.1.1.1

1.0.0.1

Figure 19: Network – Management IPv4

Parameter	Description	Mandatory	Default value
Management IPv4			
IP Address	IPv4 address for management interface	No	10.0.0.120
NetMask	IPv4 subnet mask for management interface	No	255.255.255.0
Gateway	IPv4 address default gateway for management interface	No	0.0.0.0
DNS Servers	IPv4 DNS Servers address 1 and 2	No	1.1.1.1 1.0.0.1

Alternative Local IP

Alternative local IP settings

☒ Enable alternative local IP

Alt. local IPv4 address

169.254.1.1

Figure 20: Network – Alternative IPv4 Address

When this setting is enabled (enabled by default), a 169.254.1.1 IP address will be added on the data bridge. This IP is only accessible via the wired port and cannot be accessed over the wireless link. This setting can be disabled, and you can also change the IP address to another value in the 169.254.0.0/16 range.

This IP is similar to the data bridge static IP, except it's only accessible over the wired ports, and can be set even when management VLAN is not enabled.

In the case below, both management VLAN and data VLAN are enabled, as well as data bridge static IP and alternative local IP. You can see on the dashboard how the IPs are assigned to the management and data bridges.

Management		Mgmt Vlan		Data Bridge	
Active network					
Mgmt Vlan ID	106				
IPv4 Gateway	192.168.106.1	Management IP, on the VLAN 106 network			
Address	192.168.106.163	Netmask	255.255.255.0		
Address	fe80::7a5e:e8ff:fed0:12a / 64				

Figure 21: Network – Management IP

Network Information

Management		Mgmt Vlan		Data Bridge	
Active network					
Bridge MAC	C4:93:00:4F:EF:4A				
DNS Servers	8.8.8.8				
IPv4 Gateway	-				
IPv6 Gateway	-				
Data Vlan ID	99				
Address	169.254.1.2	Netmask	255.255.0.0	Alternative local IP	
Address	192.168.202.111	Netmask	255.255.255.0	Data bridge static IP	
Address	fe80::7a5e:e8ff:fed0:12a / 64				

Figure 22: Network – Alternative Local IP

Management IPv6

When Management IPv4 mode is set to static, you need to provide an IP address, a gateway and DNS Servers to the device. You could also use IPv4 instead of IPv6 or both.

Management IPv6

☒ Enable IPv6 static IP

IP address: Prefix:

Gateway:

DNS servers:

Figure 23: Network – Management IPv6

Parameter	Description	Mandatory	Default value
Management IPv6			
IP Address	IPv6 Address for management interface	No	2001::1
Prefix Length	Number of bits used by the prefix	No	64
Gateway	IPv6 address default gateway for management interface	No	::
DNS Servers	IPv6 DNS Servers address 1 and 2	No	2606:4700:4700::1111 2606:4700:4700::1001

6.2.2 Interface Settings

The screenshot displays the 'Interface settings' configuration page. It features two main sections: 'General' and 'Ethernet port #1'. The 'General' section contains three input fields: 'MTU' with the value 1500, 'Bridge ageing time (sec)' with the value 300, and 'Max FDB entries' with the value 0. Below these is a toggle switch for 'Enable data VLAN'. The 'Ethernet port #1' section contains three toggle switches: 'Enable PoE out', 'Disable link data', and 'Enable failover to external device'. The top navigation bar shows 'Device management', 'Interface settings' (selected), and 'Advanced settings'.

Figure 24: Network Settings – Interface settings

MTU: Maximum transmission unit. This setting will apply to the wireless interfaces, ethernet ports, and management/local bridge. The minimum allowed value is 1280, and the max is 7900.

Bridge ageing time: Ageing determines the number of seconds a MAC address is kept in the FDB after a packet has been received from that address. Set to this 0 to disable ageing.

Max FDB entries (Station only): Set the maximum number of allowed FDB (forwarding database) entries allowed. Set this to 0 for an unlimited number (default). This will effectively limit the number of MAC addresses that can pass traffic via the ETH1 port. This is useful for limiting the number of connected customer devices - for example, set this value to 1 to ensure only one customer router is plugged in.



Max FDB entries:

- This setting is only visible when operating in Station mode.
- This setting is not compatible with failover mode.

Enable Data VLAN (Station only): When Data VLAN is enabled, traffic with the specified VLAN ID received over the upstream wireless link will have the VLAN tag removed as it exits the wired ports. Similarly, traffic coming into the device over the wired ports will be tagged with the specified VLAN ID when it's sent over the wireless link. By default, VLAN ID is set to 99.



Enable Data VLAN is available only on the Station.

The local web UI will still be accessible from the wired ports when data VLAN is enabled unless management VLAN is enabled.

Ethernet Port #1 Settings

Ethernet port #1

☒ Enable PoE out

☒ Disable link data

☒ Enable failover to external device

RSSI threshold
 -68

Flap protection time (sec)
 10

Figure 25: Network – ETH1 settings

Enable PoE Out: Enable or disable PoE out on the ETH1 1G interface. Please refer to the Operating Manual for more information about input and output power.

Disable link data: Disable data over the ETH1 interface. This is useful if you need to power an external device via PoE out but do not want to allow data to pass over the link. Note: When data is disabled over ETH1, failover will no longer function.

Enable Failover: (Station only, and when data VLAN is disabled). Enable failover/backup over a device connected to the ETH1 port. When Failover is activated, the device will use an alternative upstream data path via a device attached to the eth1 port.

Instead of relying on a low-quality link, you can use whatever radio you'd like - it can even be pointed to a completely different site.



Make sure to enable failover before you connect the network cable to your failover device connected to ETH1 to prevent network loops.

RSSI Threshold: The wireless signal threshold at or below which the device will switch into failover mode. Default value is -68 dBm, you could set a value between -66 and -70 dBm.

Flap Protection Time: The amount of time in seconds to wait before switching back to normal operation once the wireless link RSSI has moved back into the normal range. Once this happens and the flap timer countdown has begun, the flap timer will be reset if at any time the RSSI drops back below the threshold during the countdown. Flap protection could be set between 5 and 1800 seconds. Default value is set to 10 seconds.

Failover Behavior

When Failover is configured the alternative data path will be used under the following circumstances:

- The station is not currently associated to an AP, or
- The RSSI of the wireless link drops below the pre-configured RSSI threshold (see the web UI guide for more details about these settings).

The device will switch back to the wireless uplink data path once two conditions are met:

- The station is connected to the access point at an RSSI that's above the pre-configured failover threshold, and
- The RSSI of the link has been above the failover threshold for a time defined in the "Flap protection time (sec)" field under the failover settings.

Once the device switches to Failover, the wireless signal is checked every second to trigger a switch back.

The current failover status is accessible through the Ethernet Status section of the Dashboard. It will be updated once the device's wireless signal is back within a normal range, and the flap timer is counting down.

Ethernet Status		Internal Switch MAC C4:93:00:4F:EF:4A	
ETH0 2.5G Port		ETH1 PoE Out Disabled 1G Port	
Status	Bytes (Tx/Rx)	Status	Bytes (Tx/Rx)
Link up	46.8 MB/4.6 MB	Link down	-
Speed	Packets (Tx/Rx)	Speed	Packets (Tx/Rx)
1 Gbps Full Duplex	356 k/32 k	-	-
		Failover Status ⚠ Failover is active: current signal is -55 dBm and remaining flap time is 14 seconds	

Figure 26: Network – Failover Status on Dashboard

Once the data path is switched over to the failover data path you should expect to lose ping.

There are a couple of different factors that could affect ping loss during failover state changes (like how long it takes for ARP to switch over), but generally only a few pings will be lost.

Activity events, log entries, and SNMP traps are all created whenever the device's failover state changes.

Date & Time	Message
2023-06-21 11:14	Leaving failover state, resuming normal 60GHz operation with a signal of -63
2023-06-21 11:14	prs0 is up
2023-06-21 11:14	60GHz failing over to ETH1 datapath due to no wireless link
2023-06-21 10:53	Device rebooted

Figure 27: Network – Failover Status in Activities



Failover is automatically disabled if there is no active connection on the ETH1 port. This will be indicated by a message at the top of the Dashboard as well as under the ETH1 Status.

Ethernet Status		Internal Switch MAC C4:93:00:4F:EF:4A	
ETH0 2.5G Port		ETH1 PoE Out Disabled 1G Port	
Status	Bytes (Tx/Rx)	Status	Bytes (Tx/Rx)
Link up	46.8 MB/4.6 MB	Link down	-
Speed	Packets (Tx/Rx)	Speed	Packets (Tx/Rx)
1 Gbps Full Duplex	356 k/32 k	-	-
		Failover Status ⚠ Failover is disabled: No ETH1 link detected	

Figure 28: Network – Failover Disabled on Dashboard



The service interruption between the switching is 2 to 5 seconds. Some packets can be lost during the failover operation.

6.2.3 Advanced Settings

Figure 29: Network – Advanced Settings

DHCP Snooping



These settings are visible only when operating in Station or PtP slave modes.

DHCP Snooping

- ☒ Block rogue DHCP servers
- ☒ Enable DHCP Option 82 Injection

Option 82 Remote ID Type

Custom field

Remote ID Custom Value

Must be 1-64 symbols long

Option 82 Circuit ID Type

Station's wireless MAC address

Figure 30: Network – DHCP Snooping

Block rogue DHCP servers: When this setting is enabled, DHCP discovery packets are dropped at Station side before passing to downstream* and 'DHCP offer' packets are dropped at Station side before passing to upstream*. Rogue DHCP servers can occur when a user plugs their router in backwards, exposing the DHCP server to the upstream* WAN network, instead of their local network.

Enable DHCP Option 82 Injection: Enable this setting to inject DHCP Option 82 field into the upstream* DHCP request packets. You can choose to populate the Remote ID field, the Circuit ID field, or both. These fields can be set to one of the following options:

Station's wireless MAC: Insert the Station wireless MAC address into the specified DHCP option 82 field. When the MAC address is inserted, it will be ASCII encoded, and will include the colons. If the "append null terminator" option is selected, a trailing null terminator character will be attached to the end of the string. The station MAC, 78:5e:e8:d0:00:02 would show up on the DHCP server as 37:38:3A:35:65:3A:65:38:3A:64:30:3A:30:30:3A:30:32:0 with the null terminator, and 37:38:3A:35:65:3A:65:38:3A:64:30:3A:30:30:3A:30:32 without it.

Custom: Insert an ASCII string of your choice (such as a customer's ID number or phone number) into the specified option 82 field. The string must be between 1 and 64 characters. The string will be hex encoded and include a null terminator character at the end if the "append null terminator" setting is enabled.

For example, the string "terranet" will appear on the DHCP server as 74:65:72:72:61:6E:65:74:0 with the null terminator appended, or 74:65:72:72:61:6E:65:74 without it.

Additional options:

Append null terminator: this option allows to add or remove a trailing 0x0 character from the DHCP option string. When the option to append the eth port (see below) is enabled, no null terminator will be appended to the string. You may want to disable this setting if your DHCP server isn't parsing the DHCP option value correctly.

Append eth port: When this option is enabled, a label will be appended to the end of the remote or circuit ID indicating which ethernet port the client is connected behind (eth0 or eth1). For example, if the circuit ID is set to "terranet vxx ", the resulting ID in the DHCP lease for a client connected behind eth1 will show up as "terranet vxx eth1".

None: don't insert anything into the specified option 82 field

Any DHCP requests that come from connected devices over the wired ports of the remote Station, as well as from the Station itself, will be injected with the DHCP option 82 fields.



* *Upstream* indicates traffic over the uplink, from the Station to the Access point.

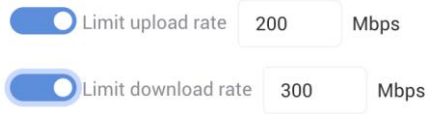
Downstream indicates traffic originating at the station, moving downstream to any devices connected to the wired ports.

Traffic Control



These settings are visible only when operating in Station or PtP slave modes.

Traffic Control

The interface shows two settings for traffic control. The first setting is "Limit upload rate", which has a blue toggle switch turned on, a text input field containing the value "200", and the unit "Mbps" to its right. The second setting is "Limit download rate", which also has a blue toggle switch turned on, a text input field containing the value "300", and the unit "Mbps" to its right.

☒ Limit upload rate Mbps

☒ Limit download rate Mbps

Figure 31: Network – Traffic Control

Limit upload rate: Enable or disable traffic shaping on the upload (wireless) path and set upload or download limit in Mbps.

Limit download rate: Enable or disable traffic shaping on the download (ETHx) path and set upload or download limit in Mbps.

6.3 Wireless Settings



Making any changes to wireless settings will cause a full networking restart, and all wireless peers will be disconnected.

Configuration > Wireless

Network **Wireless** Services System

60 GHz Radio

Wireless mode

Access point

Antenna kit

Base unit only (no antenna kit)

Channel width

Full: 2.16 GHz

Channel

1 (58320 MHz)

Max MCS

MCS 12

SSID

radwin-ptmp2

Security mode

AES+GCMP

Passphrase

Figure 32: Wireless Settings – AP

60 GHz Radio

Wireless mode
Station

Channel width
Full: 2.16 GHz

Max MCS
MCS 12

Failover
Failover to ETH1 is **not enabled**.

Station Profiles ?

☒ Enabled

Priority	SSID	Security Mode	Security Passphrase
1	RADWIN-Test	AES+GCMP	*****

☒ Enable sorting

Figure 33: Wireless Settings – Station

Wireless Mode: Choose whether you would like your radio device to operate as an Access Point (AP), Station, Point-to-Point Access Point, or Point-to-Point Station.

In Point-to-Point Access Point mode, only a single Point-to-Point Station connection will be allowed, and the link is optimized for PtP performance.



If you change the Wireless Mode, the device will require a reboot to apply the new configuration.

Antenna Kit: RADWIN TerraNet V90 could come with multiple antenna lense options. To allow the device to correctly calibrate with the antenna, you need to select the one you are using or “Base unit only” if you are not using any external antenna.

Antenna kit

Base unit only (no antenna kit) ▾

- Base unit only (no antenna kit)
- TN PTP 33dBi: 100mm/4in antenna kit, 33 dBi
- TN PTP 37dBi: 150mm/6in antenna kit, 37 dBi
- TN PTMP 45: 45° scanning antenna kit, 30 dBi
- TN PTMP 90: 90° scanning antenna kit, 30 dBi

Figure 34: Wireless Settings – Antenna Selection

Channel Width: Full (2.16 GHz) or Half (1.08 GHz). See notes below about half channel support.



If you change Channel Width, the device will require a reboot to apply the new configuration.

Here is the list of available channels in TerraNet based on the Channel Width configuration:

60 GHz Full Band Channels (2.16 GHz Bandwidth)

Channel Number	Center Frequency (GHz)	Lower Frequency (GHz)	Upper Frequency (GHz)
1	58.32	57.24	59.40
2	60.48	59.40	61.56
3	62.64	61.56	63.72
4	64.80	63.72	65.88
5	66.96	65.88	68.04
6	69.12	68.04	70.20

60 GHz Band – 13 Channels (1.08 GHz Bandwidth)

Channel Number	Center Frequency (GHz)	Lower Frequency (GHz)	Upper Frequency (GHz)
0	57.78	57.24	58.32
1-Low	58.86	58.32	59.40
1-High	59.94	59.40	60.48
2-Low	61.02	60.48	61.56
2-High	62.10	61.56	62.64
3-Low	63.18	62.64	63.72
3-High	64.26	63.72	64.80
4-Low	65.34	64.80	65.88
4-High	66.42	65.88	66.96
5-Low	67.50	66.96	68.04

5-High	68.58	68.04	69.12
6-Low	69.66	69.12	70.20
6-High	70.74	70.20	71.28*

* Only allowed in regions that license up to 71 GHz, such as the U.S. or select other countries. Not allowed in ETSI, Japan, or South Korea.

Channel (AP only): Select the desired operating channel. The available channels for the full channel width are 1-6 (for UNI version) or 1-4 (for LIM version), and for the half channel width: 1-11 (for UNI version) or 1-7 (for LIM version).



The value in parentheses following the channel number indicates the lowest frequency within the selected channel's bandwidth.

Max MCS: Data rates are dynamically selected, but you can choose to set the maximum transmit MCS at each radio device. Limiting Max MCS may be useful for achieving maximum link quality and robustness.



When half channel support is enabled, the max MCS allowed is MCS 9.

Setting max MCS only affects the TX MCS rate of the current device. Adjusting Max MCS on an Access Point will have an impact on Downlink performance for all connected Stations, while adjusting Max MCS on a Station will affect Uplink performance for the specific Station.

SSID: The radio's SSID/network name.

Security mode: Select link encryption mode.

- **AES+GCMP** is the default and recommended security mode, which is using 128-bit encryption key aligned with the WPA2 standard requirements
- **Open** mode is also available. In this mode, Station authentication and data encryption are disabled

Passphrase: When Security Mode is AES+GCMP, you will need to define a passphrase to secure the link. The default value is 'passphrase' and it is highly recommended to change it to ensure network security.

6.3.1 Station profiles

When your device is operating in Station (Client) or Point-to-Point Slave mode, an additional option is shown in the UI that allows you to enable Station Profiles. You can then define in the interface multiple connection profiles.

The client will connect to the SSID/profile with the highest priority first.

Station Profiles ?

☒ Enabled

Priority	SSID	Security Mode	Security Passphrase
1	network-23B-A	Open	
2	network-23B-B	AES+GCMP	
10	default	AES+GCMP	

+ Add

☐ Enable sorting

Figure 35: Wireless Settings – Station Profiles

For each profile you will need to define different parameters:

Priority: The priority of the profile. 1 is the highest priority, and 10 is the lowest. When you define multiple profiles with the same priority, the device will connect to the SSID that has the better signal.

SSID: The SSID for the connection profile.

Security Mode and Passphrase: Security mode and passphrase that should be used when connecting to the specified AP.

Enable sorting: Check this box to disable sorting of the profiles table while adding profiles to avoid entries reshuffling based on defined profile priorities.

6.4 Services Settings

In the Services tab, you can configure several services:

- HTTP Server.
- NTP.
- Remote Syslog.
- SNMP Traps.
- SSH Server.
- Device Discovery.
- SNMP Server.

- Ping Watchdog.

Network
Wireless
Services
System

HTTP server

Configure the ports used to access this device's local web server.

Port

HTTPS port

SSH Server

Secure Shell Protocol (SSH) is a cryptographic network protocol for operating network services securely over an unsecured network.

☒ Enabled

Port

NTP

Network Time Protocol (NTP) is a protocol used to synchronize this device's system clock time.

☒ Enabled

Server addresses

Device discovery

This feature enables this device to find other devices compatible with the available discovery protocols, as well as to broadcast information to other devices.

☒ Enabled

Discover nearby devices:

☒ LLDP listener server

Broadcast device info:

☒ LLDP (Link Layer Discovery Protocol)

☒ CDP (Cisco Discovery Protocol)

☒ MNDP (MikroTik Neighbor Discovery Protocol)

Remote syslog

Syslog is a way for this device to send event messages to a logging server or file.

☐ Disabled

SNMP Server

Simple Network Management Protocol (SNMP) is an application-layer protocol for monitoring and managing network devices on a local area network (LAN) or wide area network (WAN). The purpose of SNMP is to provide network devices such as routers, servers and printers with a common language for sharing information with a network management system (NMS).

☒ Enabled

Protocol

Community

SNMP Traps

An asynchronous alert sent by the local SNMP agent to the SNMP server specified below to indicate a significant event has occurred. Traps match up to system activity events for the most part.

☐ Disabled

Ping watchdog

Enable ping watchdog to reboot this device when it is unable to ping the specified IP address.

☐ Disabled

Figure 36: Configuration - Services

6.4.1 HTTP

This section refers to configuration of the local Web Server running at the device.



The Web Server is always accessible through both HTTP and HTTPS.

HTTP server

Configure the ports used to access this device's local web server.

Port

80

HTTPS port

443

Figure 37: Services – HTTP Server

Port: HTTP port at which you can access the local web UI. Default is 80.

HTTPS port: HTTPS port at which you can access the local web UI. Default is 443.



The SSL certificate for the device's web server is dynamically generated self-signed certificate.

Some modern web browsers (such as Chrome) no longer accept self-signed SSL certificates by default – in this case you will need to force access to the web server. Otherwise, you can use a browser that accepts self-signed certificates, such as FireFox.

6.4.2 NTP

NTP

Network Time Protocol (NTP) is a protocol used to synchronize this device's system clock time.

☒ Enabled

Server addresses

time.google.com

time.cloudflare.com

Figure 38: Services – NTP

Enable: Enable or disable the NTP (network time protocol) server. NTP is enabled by default.

Server addresses: A list of NTP peers that a device should use when updating the local time. By default, time.google.com and time.cloudflare.com are used.

6.4.3 Remote syslog

Remote syslog

Syslog is a way for this device to send event messages to a logging server or file.

☒ Enabled

Protocol

Please select a value

Server address

Invalid hostname

Port

Must be in range from 1 to 65535

Figure 39: Services – Remote Syslog

Enable: Enable or disable the remote syslog service.

Protocol: Remote syslog server protocol: TCP or UDP

Server address: IP address or hostname of the remote syslog server.

Port: Port at which the remote syslog server is running.

6.4.4 SNMP traps

For SNMP Traps you need to choose between SNMP v2 and SNMP v3.

SNMP Traps

An asynchronous alert sent by the local SNMP agent to the SNMP server specified below to indicate a significant event has occurred. Traps match up to system activity events for the most part.

☒ Enabled

Server address

Invalid hostname

User

Must be 1-100 symbols long

Protocol

SNMPv2

Community

public

Figure 40: Services – SNMP Traps v2

SNMP Traps

An asynchronous alert sent by the local SNMP agent to the SNMP server specified below to indicate a significant event has occurred. Traps match up to system activity events for the most part.

☒ Enabled

Server address

Invalid hostname

User

Must be 1-100 symbols long

Protocol

SNMPv3

Password

Must be 8-32 symbols long

Figure 41: Services – SNMP Traps v3

Enable: Enable or disable SNMP traps to be sent from this device.

User: The username that should be included when connecting to the server specified below. If no username is required, just use a dummy value here such as "terranet".

Server address: Hostname or IP of the SNMP trap receiver.

Protocol: Choose the trap version: SNMPv2 or SNMPv3

Community (SNMPv2 only): Community string for SNMPv2.

Password (SNMPv3 only): Password used for SNMPv3.

Once enabled, traps will be pushed to your trap server.



Because of the nature of SNMP traps, some events might be lost before they reach your trap server. For example, client association traps are sent at association time, which could occur before the upstream data path is completely initialized, causing the trap message to be dropped.

6.4.5 SSH Server

SSH Server

Secure Shell Protocol (SSH) is a cryptographic network protocol for operating network services securely over an unsecured network.

☒ Enabled

Port

22

Figure 42: Services – SSH Server

Enable: Enable or disable SSH Access to the device.

Port: Defines the port used to securely connect to the device through SSH.



Usage of SSH access should be limited to remote support activities.

6.4.6 Device discovery

Device discovery

This feature enables this device to find other devices compatible with the available discovery protocols, as well as to broadcast information to other devices.

☒ Enabled

Discover nearby devices:

☒ LLDP listener server

Broadcast device info:

☒ LLDP (Link Layer Discovery Protocol)

☒ CDP (Cisco Discovery Protocol)

☒ MNDP (MikroTik Neighbor Discovery Protocol)

Figure 43: Services – Device Discovery

Enable: Enable or disable the device discovery service for this device.



TerraNet device transmits discovery frames using CDP (Cisco Discovery Protocol).

Discover nearby devices

Enable the LLDP listener server (Link Layer Discovery Protocol): used to find nearby devices on the network. Nearby devices can be found by using the Device discovery tool on the Tools page.

Broadcast Device Info

Allow connected devices to be discoverable over different s protocols:

- LLDP (Link Layer Discovery Protocol),
- CDP (Cisco Discovery Protocol), and/or
- MNDP (Mikrotik Neighbor Discovery Protocol).

6.4.7 SNMP

When configuring SNMP, you can use different protocol combinations:

- SNMP v2
- SNMP v3
- SNMP v2 + v3

SNMP Server

Simple Network Management Protocol (SNMP) is an application-layer protocol for monitoring and managing network devices on a local area network (LAN) or wide area network (WAN). The purpose of SNMP is to provide network devices such as routers, servers and printers with a common language for sharing information with a network management system (NMS).

☒ Enabled

Protocol

SNMPv2

Community

public

Figure 44: Services – SNMP v2

SNMP Server

Simple Network Management Protocol (SNMP) is an application-layer protocol for monitoring and managing network devices on a local area network (LAN) or wide area network (WAN). The purpose of SNMP is to provide network devices such as routers, servers and printers with a common language for sharing information with a network management system (NMS).

☒ Enabled

Protocol

SNMPv3

User

Must be 1-100 symbols long

Password

Must be 8-32 symbols long

bd

Figure 45: Services – SNMP v3

SNMP Server

Simple Network Management Protocol (SNMP) is an application-layer protocol for monitoring and managing network devices on a local area network (LAN) or wide area network (WAN). The purpose of SNMP is to provide network devices such as routers, servers and printers with a common language for sharing information with a network management system (NMS).

☒ Enabled

Protocol

Community

User

Must be 1-100 symbols long

Password

Must be 8-32 symbols long

Figure 46: Services – SNMP v2 + v3

Enable: Enable the local SNMP server. The SNMP server is disabled by default.

Protocol: SNMPv2, SNMPv3, or both SNMPv2 + v3.

Community (SNMPv2 only): Input the community string for the SNMP server. The default value is public.

User (SNMPv3 only): SNMPv3 authentication username. Length must be between 1 and 100 characters.

Password (SNMPv3 only): SNMPv3 SHA+AES authentication passphrase. Length must be between 1 and 32 characters.

6.4.8 Ping watchdog

Ping watchdog service pings the specified IP address at the given interval and reboots the device after receiving a certain number of failures in a row. This service is disabled by default.

Ping watchdog

Enable ping watchdog to reboot this device when it is unable to ping the specified IP address.

☒ Enabled

Ping interval (s)

300

Startup delay (s)

300

Failure count

3

IP address(es) to ping:

192.168.1.1

Figure 47: Services – Ping Watchdog

Enable: Enable the ping watchdog service.

Ping interval: How long the service should wait, in seconds, between attempts to ping the provided IP.

Startup delay: The length of time in seconds that the service should wait until it attempts the first ping.

Failure count: The maximum allowed number of failures allowed (in a row) before the device will be rebooted.

IP address(es) to ping: The IP addresses that the service will attempt to ping. You could define just one.

6.5 System Settings

Figure 48: Configuration – System

6.5.1 Device information

Figure 49: System – Device Information

Device name: This field is used to populate the system name field used in the device discovery tool.

Device location: The physical location of this device. This free-form field is not used internally by the system and can be set to whatever you'd like.

Hostname: This must be a valid hostname format and only contain alphanumeric characters, underscores and dashes, and must start or end in an alphanumeric character.

6.5.2 Other Settings

Other settings

☒ Physical reset button enabled

Figure 50: System – Other Settings

Physical reset button: Enable or disable the physical reset button.



We do not recommend disabling the device's physical reset button. In case of misconfigurations, the device could become unreachable.

6.5.3 Time settings

Time settings

☐ Enable advanced timezones w/DST support

Time zone

(UTC+2) Europe/Athens

Date

04/10/2024



Time

04:24 PM



[Set device time from browser](#)

Figure 51: System – Time Settings

Enable advanced time zones: Select this box to enable time zones that have DST support enabled.

Time zone: The time zone that should be used for this device's time.

Date/time: Use the date and time fields to manually set the device's local date and time.



If NTP is enabled, manual set of date and time is disabled.

Set device time from browser: If the browser supports this function, the device will retrieve the date and time from the connected device.

7 Local User List

This is the default authentication method and will authenticate API and web UI users according to credentials configured in the user list. (Read more below about user roles).

Users configuration

[+ Add](#)

This is the local user database that is used for web UI authentication when the above system authentication method is set to 'Local users list' or when RADIUS is enabled but the RADIUS server is unavailable.

User name	Role	Status	Set new password
root	Admin	☒	 
Admin	Admin	☒	 
snmp	Read-Only	☒	 

Figure 52: Users

By default, only the root user is defined. You could add additional users by clicking on the **+Add** button.

For each user, you need to define these parameters:

User name: The name could contain alphanumeric characters, underscores, and dashes.

Role: Admin or Read-only.

There are currently two possible roles for a user:

- Admin: Full access to all settings in the Web UI and all RESTful APIv1 routes
- Read-only: Limited access to the Dashboard page of the web UI only, and APIv1 routes that don't affect operation of the device, such as fetching device stats.

Status: This enables or disables this user.

Set new password: Defines a new password respecting rules defined below. The goggles icon allows you to see the defined password.

- Passwords must be between 5 and 32 characters long.
- You may use the following special characters: !@#\$%^&*()?>.<~+_-/



Note

When you are logged in using a read-only user profile, the only accessible page is the Dashboard.

8 Tools

The TerraNet solution provides a set of different tools to help managing installations in PtP and PtMP configurations:

- Aiming.
- Site Survey Scan.
- Ping.
- Traceroute.
- View Log.
- Device Discovery.
- View Bridge Table.
- Sector Info.

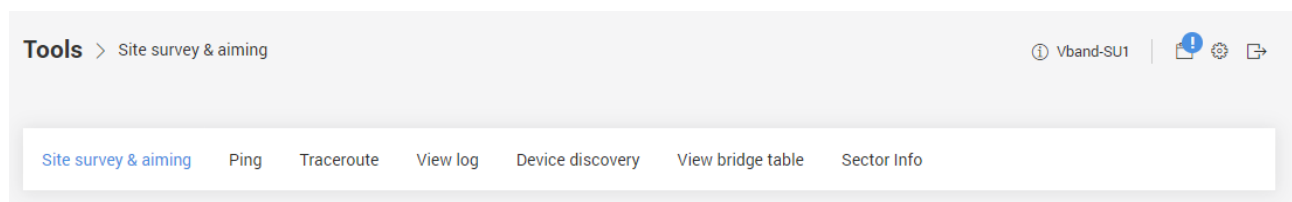


Figure 53: Tools

8.1.1 Aiming

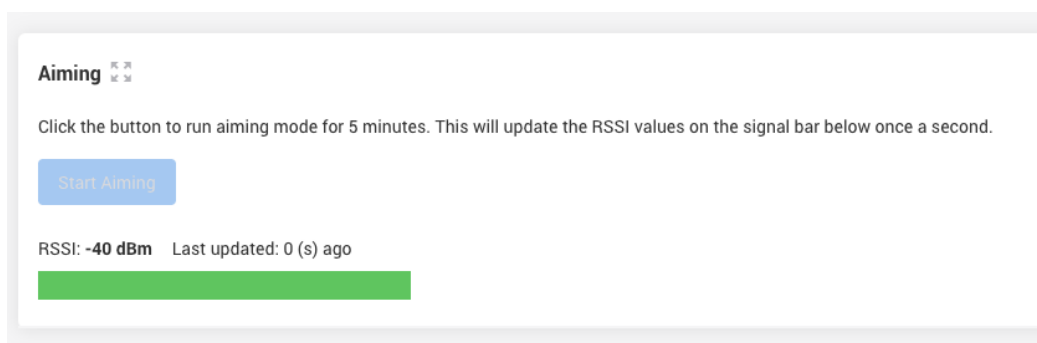


Figure 54: Tools - Aiming

During the installation phase, the Aiming tool can be used to give you RSSI signals at a faster rate than you can get it on the Dashboard. This tool is only available when operating in Station or Point-to-Point Station modes.

To view the aiming data in full screen mode, click the "Fullscreen" icon next to the Aiming header.



When installing TerraNet V90, it is recommended to use the aiming tool to optimize the alignment of the device. A wrong alignment could penalize the RSS performance by up to 10 dBm.

8.1.2 Site Survey Scan

Site survey scan

Important: This radio will go offline temporarily once scanning has started.

Select radio:

60 GHz Radio
▼

Scan

Scan results last updated: 20 hours ago

Items per page: 10 ▼

×

SSID	BSSID	Channel	Signal	Security
RADWIN-Test	78:5e:e8:00:05:c0	6 (69120 MHz)	-33 dBm	WPA2-PSK-GCMP

Total entries: 1

<<
<
1
>
>>

Figure 55: Tools – Site Survey

Use the site survey tool to view a list of other TerraNet APs broadcasting nearby. This tool is only available when operating in Station or Point-to-Point Station modes.



When running a site survey scan, the link will temporarily be disconnected. It will re-connect automatically when scanning is complete.

8.1.3 Ping

Ping tool

Use: ☒ IPv4 ☐ IPv6

IP address or host name

100.0.100.31

Ping iterations count

3

PING 100.0.100.31 (100.0.100.31): 56 data bytes

Figure 56: Tools – Ping

Perform a basic ping IPv4 or IPv6 operation from the device.

8.1.4 Traceroute

Figure 57: Tools – Traceroute

Perform a basic traceroute operation from the device.

8.1.5 View Log

Figure 58: Tools – View Log

Search and view the device's internal log. The same thread of messages is sent to remote SYSLOG servers (if defined).

8.1.6 Device Discovery

Chassis ID	Local Port	Remote Port	IPv4 address	IPv6 address	Name	VLAN
00:40:C7:5D:DF:48	eth0	2	10.0.0.121	-	IDU-S	-
Description 10-Port GbE L2+ Managed PoE Switch						
78:5E:E8:00:05:BE	prs0	br-wan	10.0.100.31	fe80::7a5e:e8ff:fe00:5be	Vband-AP1	-
Description TerraNet v40: r1.11.4 v54358						

Figure 59: Tools – Device Discovery

Use the device discovery tool to find other devices on the same local segment of the network.



You must have Device Discovery enabled under the

Configuration → Services → Device Discovery settings page for your device(s) to be discoverable.

System name and location can be set under your device's system settings located at

Configuration → System → Device information.

8.1.7 Bridge Table

MAC address	Device	Master	Port
00:40:c7:5d:df:4a	eth0	br-wan	
00:40:c7:5d:df:49	prs0	br-wan	
00:e0:4c:36:1a:64	prs0	br-wan	
78:5e:e8:00:05:be	prs0	br-wan	

Figure 60: Tools – Bridge Table

Use the bridge table tool to view MAC addresses in the device's bridge forwarding table, as well as their associated interface and internal bridge. In the example above, the last 3 MAC addresses are learned from the wireless interface prs0, on the br-wan bridge.

8.1.8 Sector Info

Wireless Peer Sector Locations

Use the sector map below to view the antenna sector(s) to which this device's peers are connected. In the case that a peer's TX sector differs from the RX sector, the peer will show up multiple times on the map.

Label options: ☒ Device name ☐ Device MAC ☐ Device IP

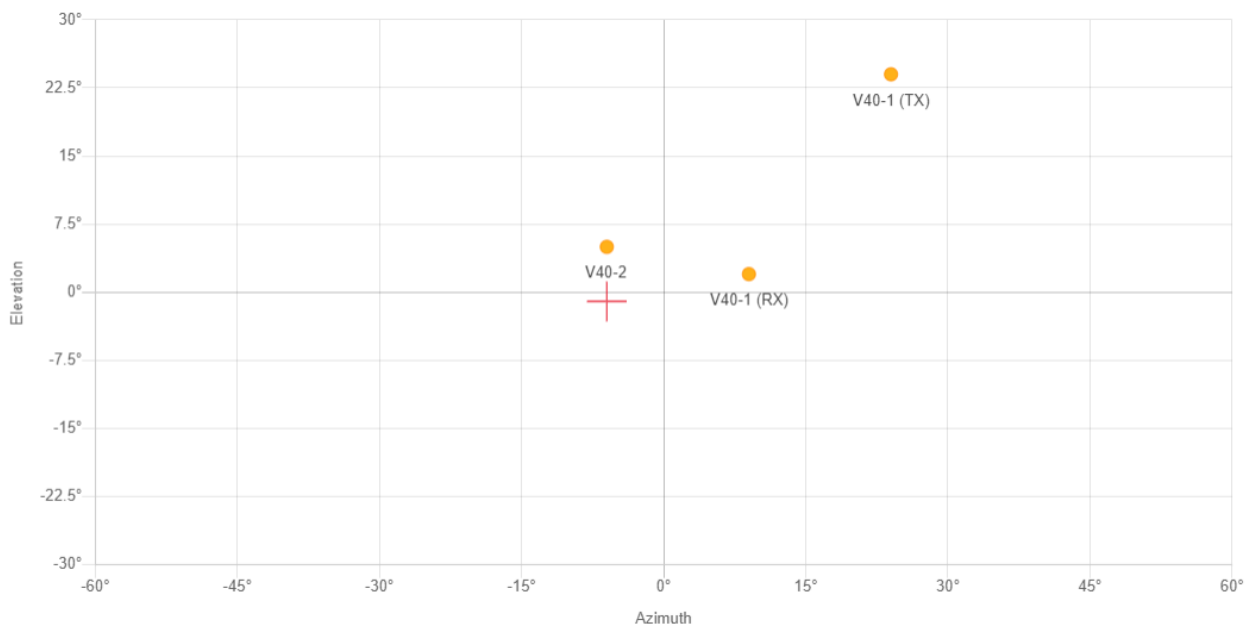


Figure 61: Tools – Sector Info

The Sector Info tool allows you to visually see the sector ID and beam azimuth/elevation for each peer (shown as an orange dot).

When you click on a connected peer's sector ID from the dashboard, you will only see the individual peer on the Sector Info tool. You can then click the "View all" button to view all connected peers.

When multiple peers are connected on the same beam / sector ID, the point will grow larger, a list of up to the first 3 devices connected on that sector will be shown.

It's possible for a peer to be connected to different sectors for RX and TX. When this happens, a peer will be represented by two dots, each with a "(RX)" or "(TX)" label following the peer's name.

9 Backups & Recovery

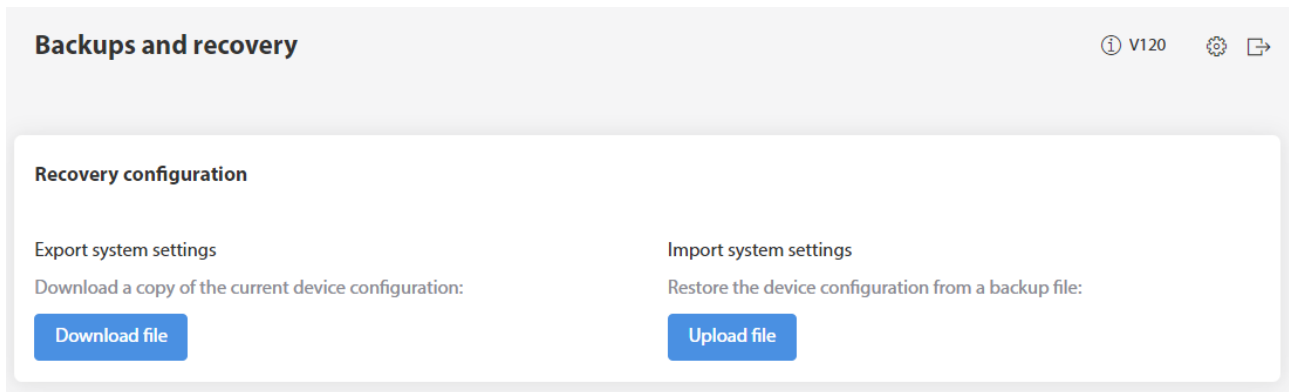


Figure 62: Tools – System Actions – Backup & Recovery

This page allows to export or import device's system settings.



You cannot restore the config of a device operating in AP mode on a device operating in Station/Client or Point-to-Point slave modes, or vice versa.

10 System Actions

You can find the system actions by clicking the gear icon located on the top right side of the page.

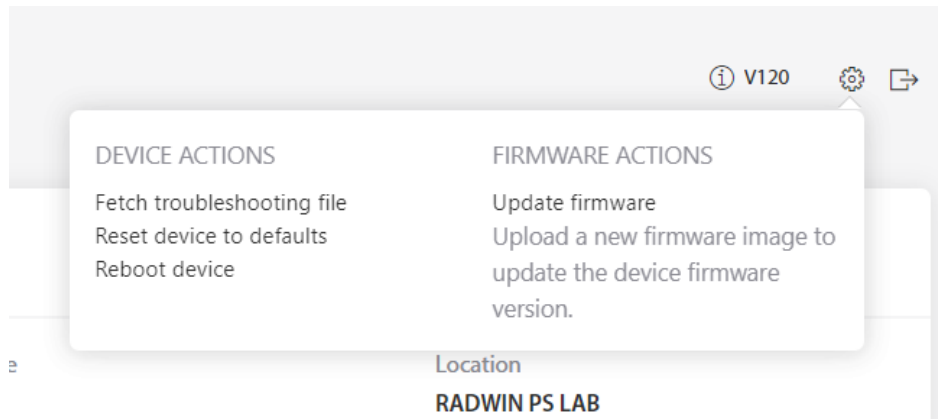


Figure 63: System Actions

10.1 Firmware Actions

10.1.1 Upgrade Firmware

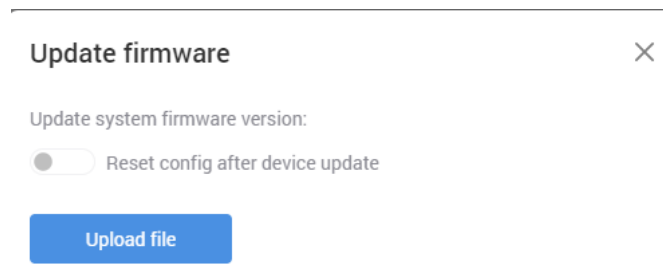


Figure 64: Tools – System Actions – Upgrade Firmware

Select this option to upgrade your device's firmware.

If after an upgrade attempt the device is running a previous version of firmware, it's possible that it failed to boot using the new firmware and fell back to the alternative boot bank (see System Information to check which firmware versions are stored in the memory banks). If there was a power interruption during the upgrade, it's safe to retry the firmware upgrade assuming the device's input power is stable.

In case you would like to align both memory banks to the same firmware version, upgrade the device twice.



Warning

Do not unplug or reboot a device while firmware upgrade is in progress!

10.2 Device Actions

10.2.1 Fetch Troubleshooting File

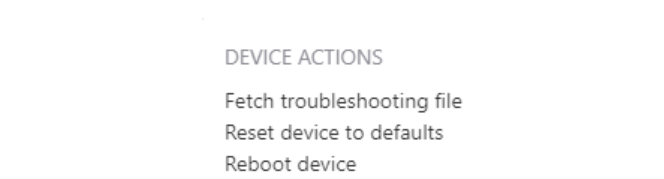


Figure 65: Tools – System Actions – Fetch Troubleshooting File

Fetch an archive of log files, configuration files, stats, and other information useful for troubleshooting with RADWIN support team.

10.2.2 Reset device to Defaults

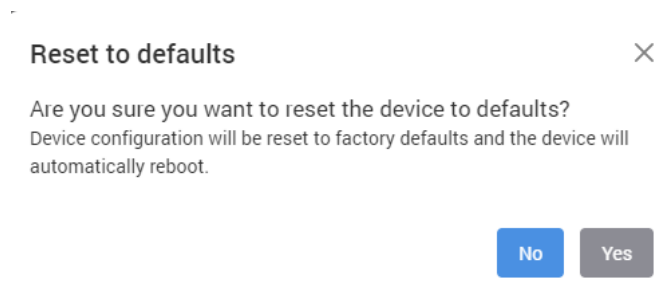


Figure 66: Tools – System Actions – Reset to Defaults

Reset your device to factory defaults (including management IP and VLAN).

10.2.3 Reboot

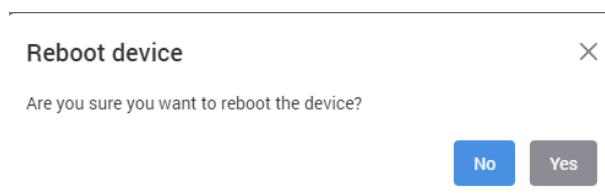


Figure 67: Tools – System Actions – Reboot

Reboot your device immediately.

10.3 Events

When events occur, an icon appears close to the gear icon.



Figure 68: Tools – System Actions – Events

When clicking on this icon, you can visualize the list of events.

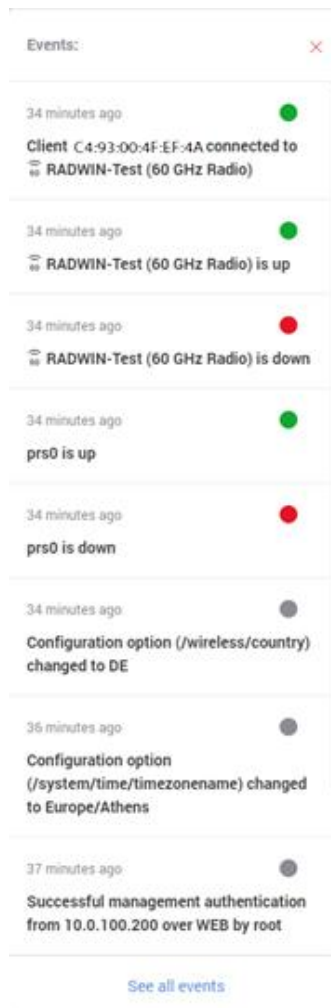


Figure 69: Tools – System Actions – List of Events

By clicking on 'See all events' you are re-directed to the activity page.

11 Appendixes

11.1 Web UI Events Table

The following events are supported in the system and displayed on the activity page.

	Event text	Comments
	Login attempt by <username> failed / succeeded	Login attempt to browser user interface
	Ethernet Service was opened / closed	Link is active (non-active) and traffic is enabled (disabled)
	Link is up	Link between AP and Client established
	Link is down due to <reason>	Link between AP and Client dropped. Possible reasons: Bad quality: link disconnected due to weak signal or high interference. User request: link disconnected due to user changing air interface configuration (e.g., CBW, Channel, Tx Ratio)
	Link state changed to <new state>	
	LAN disconnected	LAN Cable was disconnected
	LAN connected	LAN Cable was connected
	Configuration was changed	
	SW upgrade was finished successfully to unit	
	Failed to upgrade unit	

11.2 Terminology

ACRONYM	DEFINITION
AES	Advanced Encryption Standard
ARP	Address Resolution Protocol
DL	Download
EIRP	Effective Isotropic Radiated Power
IP	Internet Protocol
LAN	Local Area Network
LLDP	Link Layer Discovery Protocol
MCS	Modulation Coding Scheme
POE	Power Over Ethernet
ETH	Ethernet
SNMP	Simple Network Management Protocol
SSID	Service Set Identifier
TDD	Time-Division Duplex
TX	Transmit
RX	Receive
UL	Upload
VLAN	Virtual Local Area Network

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